Trabalho 6 e 7 - LS-MRAC e Combined M-MRAC + LS

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Professor:

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LS-MRAC

Resumo do Algoritmo

Subsystem	Equation	Order
Plant	$y = P(s)\,u$	n
Model	$y_m = M(s)\,r$	n
Track. error	$e_a = \mathrm{sign}(k_p)(y - y_m)$	
SV Filters	$\dot{\omega}_1 = A_f\omega_1 + b_f u$ $\dot{\omega}_2 = A_f\omega_2 + b_f y$	$n-1$ $n-1$
Regressor	$\omega^T = \begin{bmatrix}\omega_1^T & y & \omega_2^T & r\end{bmatrix}$	
ξ-Filter	$\xi = L^{-1}(s)[\omega]$	$2n$
Control law	$u = \theta^T\omega + \dot{\theta}^T\xi$	
Update law	$\dot{\theta} = -\gamma\, R\xi e_a\,, \quad \gamma > \dfrac{1}{2\|k_p\|}$ $\dot{R} = -R\xi\xi^T R\,,$ $R(0) = R^T(0) > 0$	$2n$ $4n^2$

Análise de Estabilidade

Partindo do M-MRAC, temos:

Lei de controle: $\boxed{u = L(s)\theta^T\xi} \rightarrow \boxed{u = \theta^T\omega + \dot{\theta}^T\xi}$

com, $\boxed{L(s) = s + \ell_0}$, $\ell_0 > 0$ e $\boxed{\xi = L^{-1}(s)\omega}$

Erro: $e_a = |k_p|M(s)\left[L(s)\theta^T\xi - \theta^{*T}\omega\right] = |k_p|\underbrace{M(s)L(s)}_{n^*=0}\left[\tilde{\theta}^T\xi\right]$

$M(s)L(s)$ Pode ser decomposto como:

$$M(s)L(s) = \frac{s+\ell_0}{s+a_m} = \frac{\ell_0 - a_m}{\underbrace{s+a_m}_{SPR \text{ if } \ell_0 > a_m}} + 1 = \alpha M(s) + 1$$

Resultando em: $e_a = \underbrace{\alpha|k_p|M(s)}_{\text{SPR}}\left[\tilde{\theta}^T\xi\right] + |k_p|\left[\tilde{\theta}^T\xi\right]$,

$$\begin{cases} \dot{e} = A_m e + B_m[\tilde{\theta}^T\xi] \\ e_a = C_m e + |k_p|[\tilde{\theta}^T\xi] \end{cases} \quad (e \in \mathbb{R}^{5n-2})$$

Função de Lyapunov: $2V = e^T e + \tilde{\theta}^T\Gamma^{-1}\tilde{\theta}$

Derivando, $\dot{V} = -e^T Q e + \underbrace{e^T P B_m}_{e_a - |k_p|[\tilde{\theta}^T\xi]}\left[\tilde{\theta}^T\xi\right] + \tilde{\theta}^T\Gamma^{-1}\dot{\theta}$

Rearranjando,

$$\dot{V} = -e^T Q e - |k_p|\left(\tilde{\theta}^T\xi\right)^2 + \tilde{\theta}^T\left[\xi e_a + \Gamma^{-1}\dot{\theta}\right]$$

Adaptação: $\boxed{\dot{\theta} = -\Gamma\,\xi e_a}$

Resultado: $\boxed{\dot{V} = -e^T Q e - |k_p|\left(\tilde{\theta}^T\xi\right)^2 \leq 0}$

Análise de Estabilidade

Adicionando a lei de adaptação do LS, temos:

Função de Lyapunov: $\boxed{2V(e,\tilde{\theta}) = \gamma e^T P e + \tilde{\theta}^T R^{-1}(t)\tilde{\theta}}$ $\quad R(0) = R^T(0) > 0$

Derivando, $\dot{V} = -\gamma e^T Q e - \gamma |k_p| (\tilde{\theta}^T \xi)^2 + \gamma \tilde{\theta}^T \xi e_a + \tilde{\theta}^T R^{-1} \dot{\theta} + \frac{1}{2} \tilde{\theta}^T \dot{R}^{-1} \tilde{\theta}$

Relembre que: $\frac{d}{dt}[RR^{-1}] = \dot{R}R^{-1} + R\dot{R}^{-1} = 0 \quad \Rightarrow \quad \boxed{\dot{R}^{-1} = -R^{-1}\dot{R}R^{-1}}$

Portanto, $\dot{V} = -\gamma e^T Q e - \gamma |k_p| (\tilde{\theta}^T \xi)^2 + \tilde{\theta}^T [\gamma \xi e_a + R^{-1}\dot{\theta}] - \frac{1}{2} \tilde{\theta}^T \underbrace{R^{-1}\dot{R}R^{-1}}_{-\xi\xi^T} \tilde{\theta}$

Adaptação: $\boxed{\dot{\theta} = -\gamma R \xi e_a}$, $\boxed{R^{-1}\dot{R}R^{-1} = -\xi\xi^T}$ $\Rightarrow$ $\boxed{\dot{R} = -R\xi\xi^T R}$

Resultado: $\dot{V} = -\gamma e^T Q e - \gamma |k_p| (\tilde{\theta}^T \xi)^2 + \frac{1}{2} (\tilde{\theta}^T \xi)^2 = -\gamma e^T Q e - \Big(\underbrace{\gamma |k_p| - \frac{1}{2}}_{>0 \text{ if } \gamma > \frac{1}{2|k_p|}} \Big) (\tilde{\theta}^T \xi)^2 \le 0$

Análise de Estabilidade

Condição de estabilidade: $\boxed{\gamma > \frac{1}{2|k_p|}}$ $\Rightarrow$ $\dot{V}(e,\tilde{\theta}) \leq 0$

$$\begin{cases} \bullet\ e \in \mathcal{L}_\infty \ , \quad \tilde{\theta}^T R^{-1} \tilde{\theta} \in \mathcal{L}_\infty \\ \bullet\ e \in \mathcal{L}_2 \ , \quad \tilde{\theta}^T \xi \in \mathcal{L}_2 \\ \bullet\ \omega,\ \xi,\ \dot{\xi} \in \mathcal{L}_\infty \end{cases}$$

Como $R(0) = R^T(0) > 0$, temos que: $\dot{R}^{-1}(t) = \xi\xi^T$

Integrando: $R^{-1}(t) = R^{-1}(0) + J(t) > 0\,, \quad t \geq 0$ onde, $J(t) = \int_0^t \xi(\tau)\xi^T(\tau)d\tau$

Portanto, temos que $R^{-1}(t) > R^{-1}(0)$, dessa forma $R(t) > 0,\ \forall t \geq 0$ e $R \in \mathcal{L}_\infty$ Isso significa que: $\dot{\theta}, \dot{R} \in \mathcal{L}_\infty$

Da função de Lyapunov: $2V = \underbrace{\gamma e^T P e}_{\in \mathcal{L}_\infty} + \underbrace{\tilde{\theta}^T R^{-1}(0)\tilde{\theta}}_{\in \mathcal{L}_\infty} + \underbrace{\tilde{\theta}^T J(t)\tilde{\theta}}_{\in \mathcal{L}_\infty}$

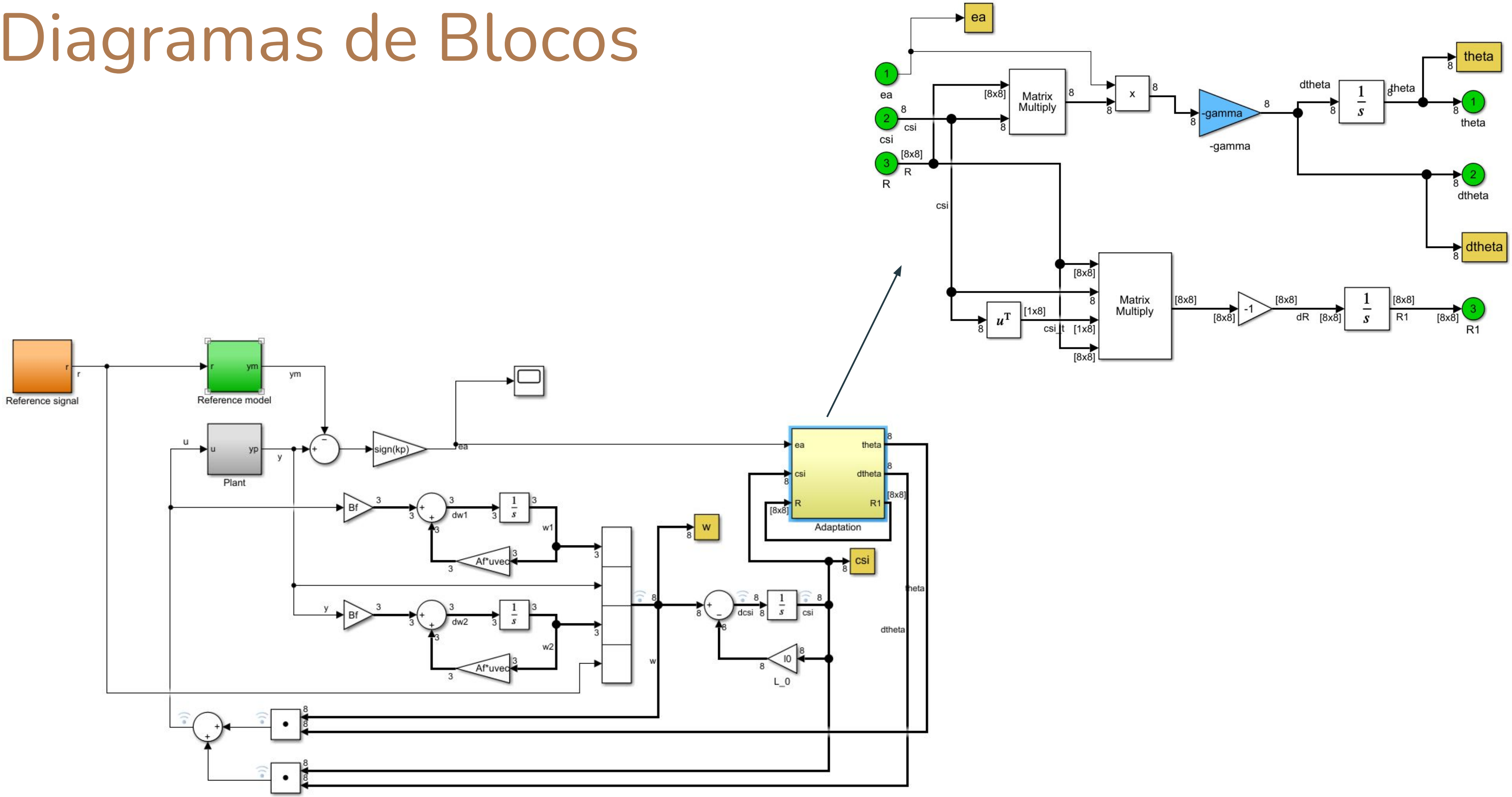
- $V \in \mathcal{L}_\infty \quad \Rightarrow \quad \tilde{\theta}^T R^{-1}(0)\tilde{\theta} \in \mathcal{L}_\infty \quad \Rightarrow \quad \tilde{\theta} \in \mathcal{L}_\infty$

- $\dot{e} \in \mathcal{L}_\infty \quad \Rightarrow \quad \lim_{t\to\infty} e(t) = 0$

- $\frac{d}{dt}\tilde{\theta}^T \xi \in \mathcal{L}_\infty \quad \Rightarrow \quad \lim_{t\to\infty} \tilde{\theta}^T \xi = 0$

- Portanto, todos os sinais são limitados $\Rightarrow$ Estabilidade uniforme global

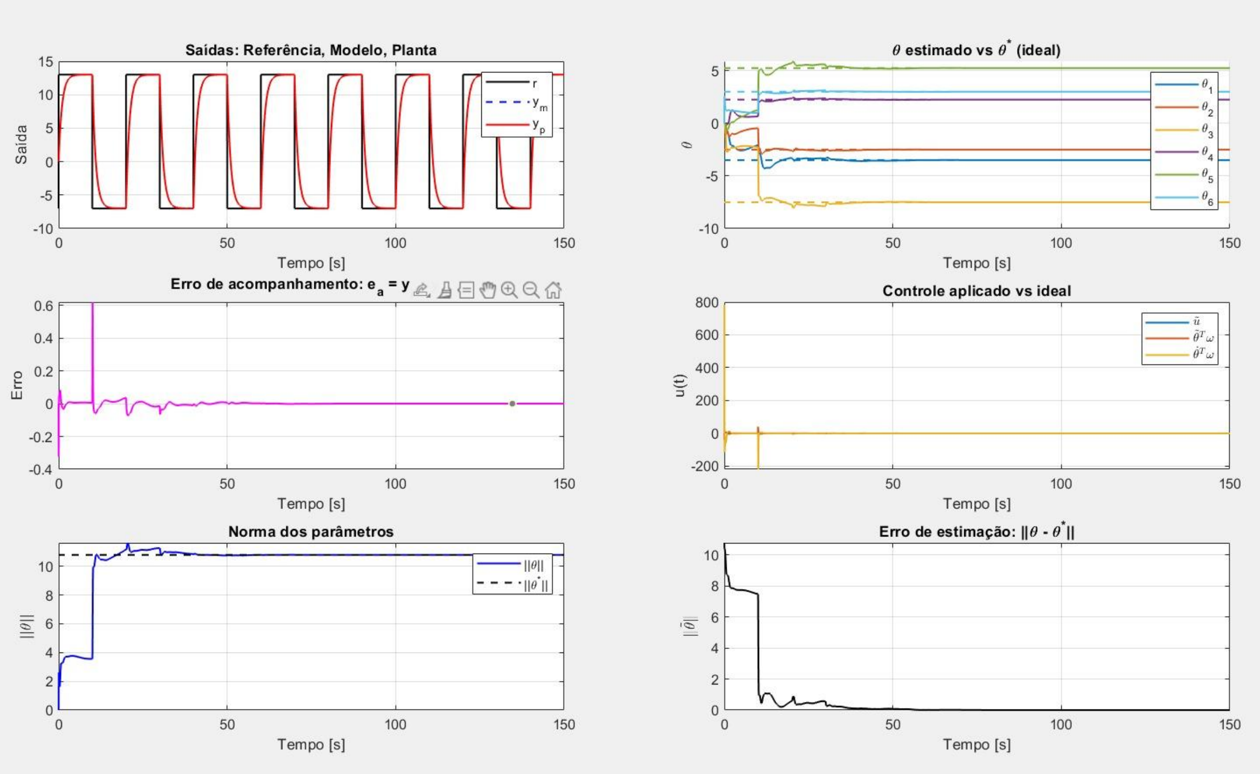
Diagramas de Blocos



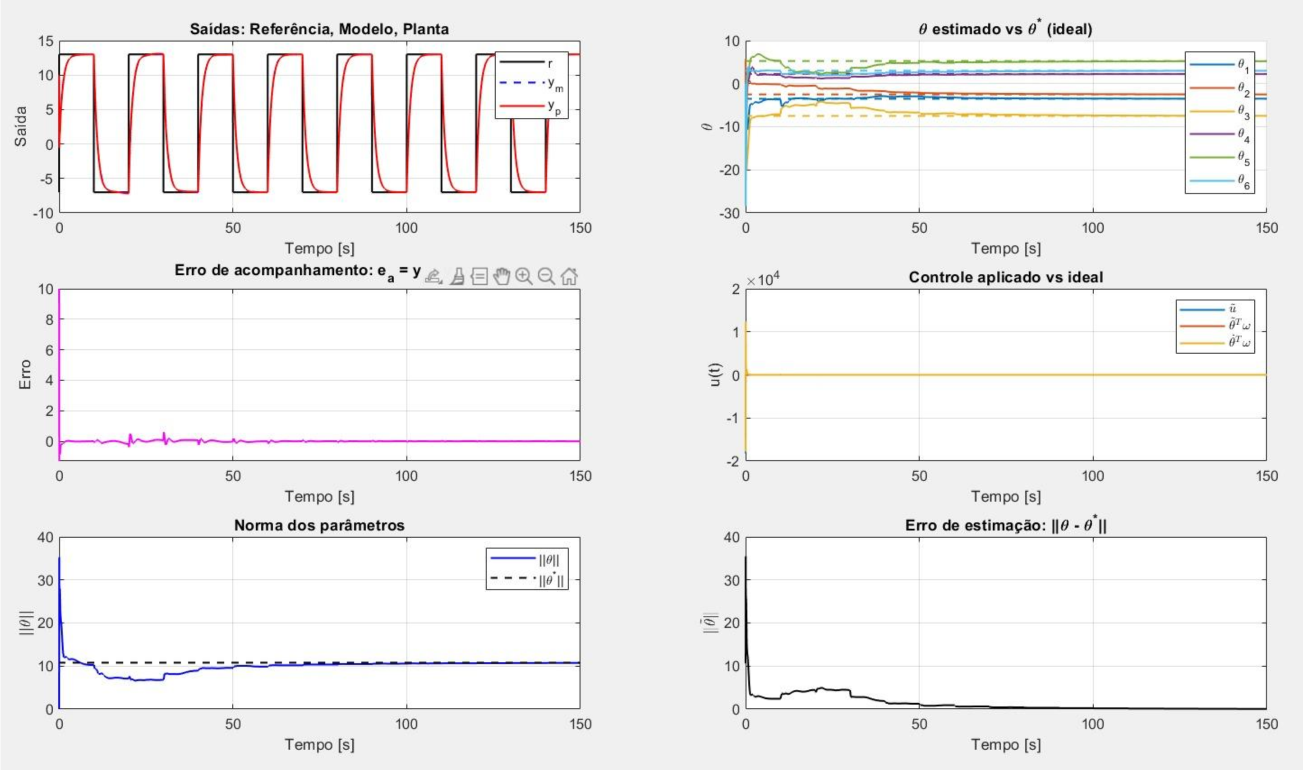
Resultados das Simulações

Base (DC = 3, SQ = 1, wq = 0.1*pi)

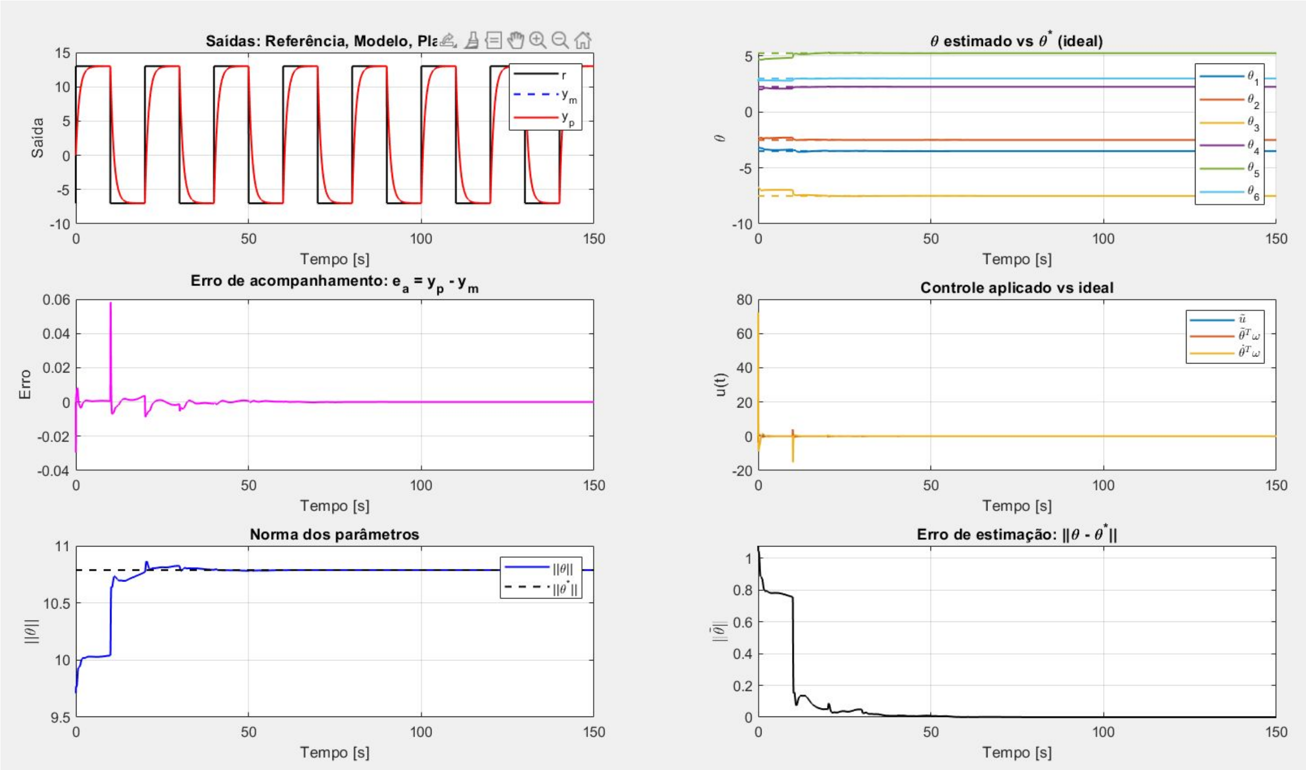
```
n = 3;  % ordem da planta e do modelo

%% ======= Inicialization =======
y0 = 0;
ym0 = 0;
gamma = 20;
R0 = 20*eye(2*n);
tfinal = 150;    %Simulation interval
st = 0.01;       %Sample time to workspace
theta0 = zeros(2*n, 1);

%% ======= Reference signal parameters =======
DC = 3;   %Constant
S1 = 0;
S2 = 0;
S3 = 0;
SQ = 1;
% Define Laplace variable
s = tf('s');
kp= 1/3;
P = kp*((s + 2)^2) / (s^3);
P = ss(P);
x0 = P.c \ y0;
M = 1 / (s + 1);
M = ss(M);
xm0 = M.c \ ym0;
Lambda = 1 / ((s + 0.5)*(s + 1)*(s + 1.5));
l0 = 2;

%% Polinômio do Observador Vetorizado
A0 = 1;
```



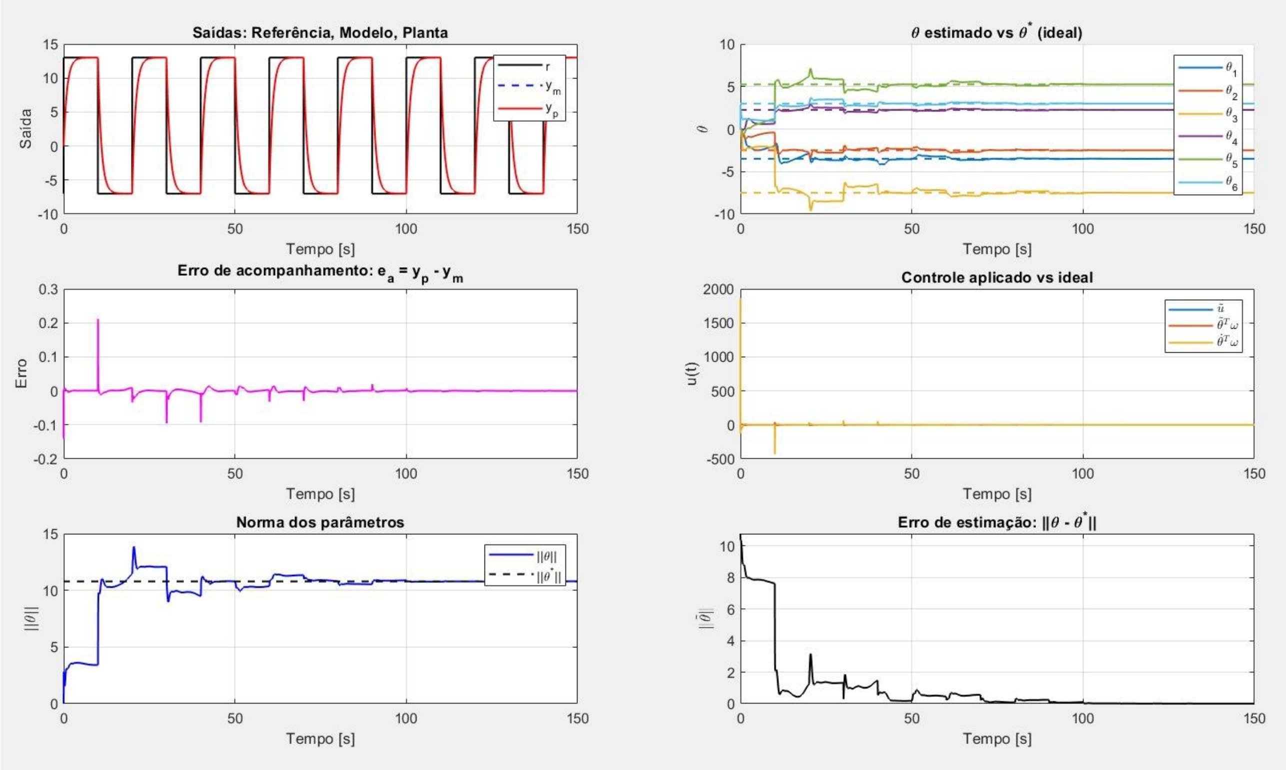
Condições Iniciais (y0 =10)



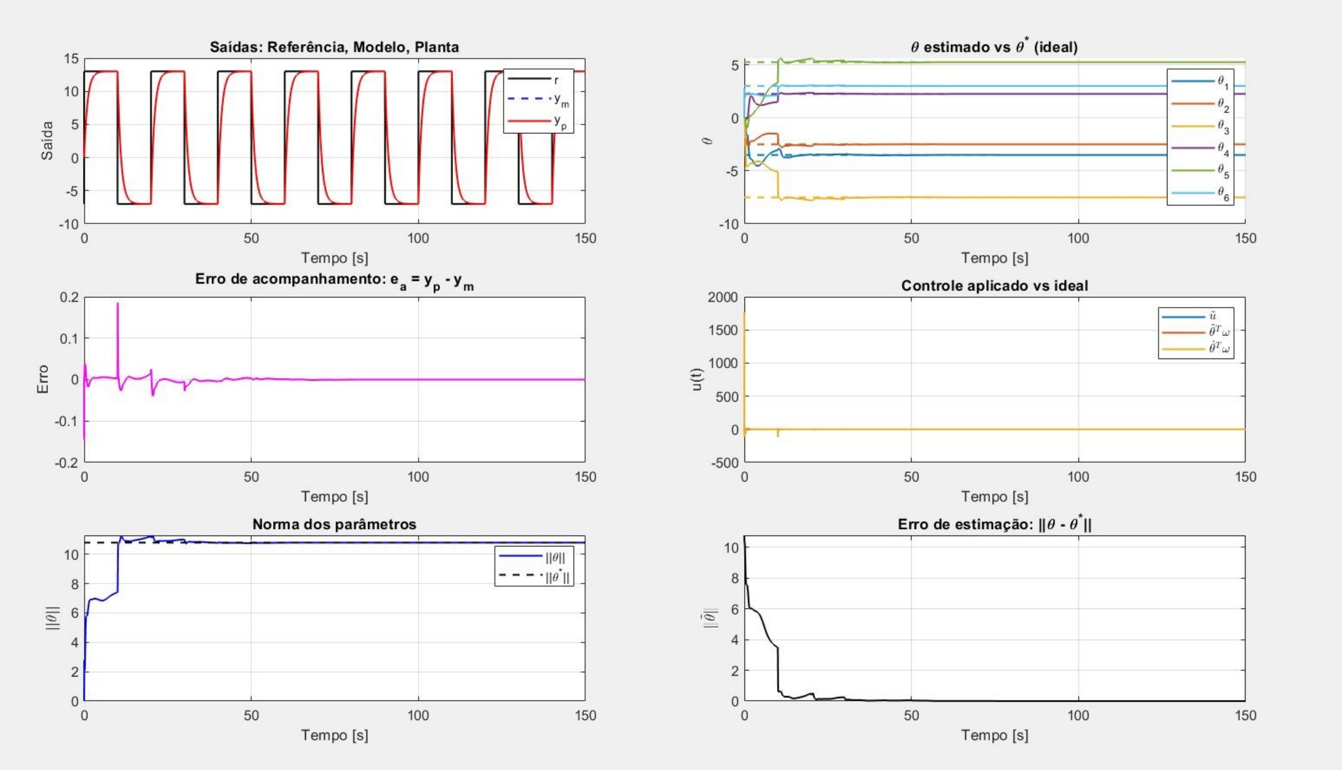
Condições Iniciais (theta = 0.9 theta star)



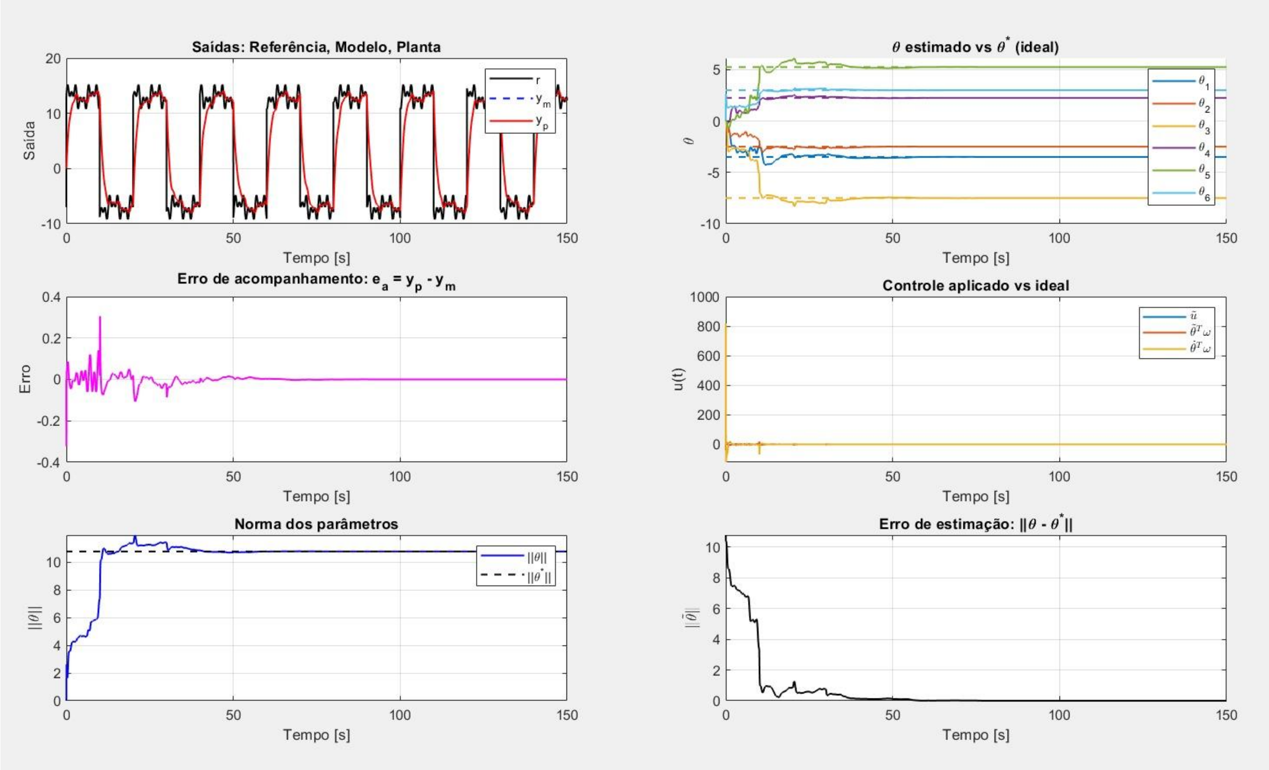
Ganho de Adaptação - gamma 200



Ganho LS - R = 200



Sinal de Referência - A = 1, w = 1/3/5 rad/s



Combined MRAC - First Order Case

Revisão do Algoritmo

$$\dot{y} = -a_p y + k_p u$$

$$\dot{y}_m = -a_m y_m + k_m r, \quad a_m > 0$$

$$e = y - y_m$$

$$\omega = \begin{bmatrix} y & r \end{bmatrix}^T$$

$$\theta = \begin{bmatrix} \theta_1 & \theta_2 \end{bmatrix}^T \qquad \theta_1^* = \frac{a_p - a_m}{k_p} \qquad \theta_2^* = \frac{k_m}{k_p}$$

$$u = \theta^T \omega$$

$$\psi = \begin{bmatrix} \psi_1 & \psi_2 \end{bmatrix}^T$$

$$\psi^* = \begin{bmatrix} a_p & k_p \end{bmatrix}^T$$

$$\dot{\hat{y}} = -a_i \epsilon - \psi_1 y + \psi_2 u, \quad a_i > 0$$

$$\epsilon = y - \hat{y}$$

$$\psi_2^* \theta_1^* - \psi_1^* + a_m = 0$$

$$\psi_2^* \theta_2^* - k_m = 0$$

$$\Delta_1 = \psi_2 \theta_1 - \psi_1 + a_m$$

$$\Delta_2 = \psi_2 \theta_2 - k_m$$

$$\dot{e} = -a_m e + k_p \tilde{\theta}^T \omega$$

$$\dot{\epsilon} = -a_i \epsilon - \tilde{\psi}_1 y + \tilde{\psi}_2 u$$

$$\dot{\theta} = -\operatorname{sign}(k_p)[\gamma_1 e \omega + \gamma_2 \Delta]$$

$$\dot{\psi}_1 = \kappa_1 \epsilon y + \kappa_2 \Delta_1$$

$$\dot{\psi}_2 = -\kappa_1 \epsilon u - \kappa_2 \theta^T \Delta$$

$$\begin{aligned} \Delta_1 &= \psi_2 \theta_1 - \psi_1 + a_m - (\psi_2^* \theta_1^* - \psi_1^* + a_m) \\ &= \psi_2 \theta_1 - (\psi_1 - \psi_1^*) - k_p(\theta_1 - \tilde{\theta}_1) \\ &= \theta_1 \tilde{\psi}_2 - \tilde{\psi}_1 + k_p \tilde{\theta}_1 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \psi_2 \theta_2 - k_m - (\psi_2^* \theta_2^* - k_m) \\ &= \psi_2 \theta_2 - k_p(\theta_2 - \tilde{\theta}_2) \\ &= \theta_2 \tilde{\psi}_2 + k_p \tilde{\theta}_2 \end{aligned}$$

Análise de Estabilidade

$$\dot{\psi}_1 = \kappa_1 \epsilon y + \kappa_2 \Delta_1, \quad \dot{\psi}_2 = -\kappa_1 \epsilon u - \kappa_2 \theta^T \Delta$$

$$V = \frac{\gamma_1}{2\gamma_2} e^2 + \frac{|k_p|}{2\gamma_2} \tilde{\theta}^T \tilde{\theta} + \frac{\kappa_1}{2\kappa_2} \epsilon^2 + \frac{1}{2\kappa_2} \tilde{\psi}^T \tilde{\psi}$$

$$\dot{V} = \frac{\gamma_1}{\gamma_2} e \dot{e} + \frac{|k_p|}{\gamma_2} \tilde{\theta}^T \dot{\theta} + \frac{\kappa_1}{\kappa_2} \epsilon \dot{\epsilon} + \frac{1}{\kappa_2} \tilde{\psi}^T \dot{\psi}$$

$$\Rightarrow \quad \frac{1}{\kappa_2} \tilde{\psi}^T \dot{\psi} = \cancel{\frac{\kappa_1}{\kappa_2} \epsilon \tilde{\psi}_1 y} + \tilde{\psi}_1 \Delta_1 - \cancel{\frac{\kappa_1}{\kappa_2} \epsilon \tilde{\psi}_2 u} - \tilde{\psi}_2 \theta^T \Delta$$

$$\dot{\epsilon} = -a_i \epsilon - \tilde{\psi}_1 y + \tilde{\psi}_2 u$$

$$\Rightarrow \quad \frac{\kappa_1}{\kappa_2} \epsilon \dot{\epsilon} = -\frac{\kappa_1}{\kappa_2} a_i \epsilon^2 - \cancel{\frac{\kappa_1}{\kappa_2} \epsilon \tilde{\psi}_1 y} + \cancel{\frac{\kappa_1}{\kappa_2} \epsilon \tilde{\psi}_2 u}$$

$$\dot{e} = -a_m e + k_p \tilde{\theta}^T \omega$$

$$\dot{\theta} = -\operatorname{sign}(k_p)\left[\gamma_1 e \omega + \gamma_2 \Delta\right]$$

$$\Rightarrow \quad \frac{\gamma_1}{\gamma_2} e \dot{e} = -\frac{\gamma_1}{\gamma_2} a_m e^2 + \cancel{\frac{\gamma_1}{\gamma_2} k_p e \tilde{\theta}^T \omega}$$

$$\Rightarrow \quad \frac{|k_p|}{\gamma_2} \tilde{\theta}^T \dot{\theta} = -\cancel{\frac{\gamma_1}{\gamma_2} k_p e \tilde{\theta}^T \omega} - k_p \tilde{\theta}^T \Delta$$

$$-k_p \tilde{\theta}^T \Delta + \tilde{\psi}_1 \Delta_1 - \tilde{\psi}_2 \theta^T \Delta = -\Delta^T \Delta \quad \Longrightarrow \quad \dot{V} = -\frac{\gamma_1}{\gamma_2} a_m e^2 - \frac{\kappa_1}{\kappa_2} a_i \epsilon^2 - \Delta^T \Delta \leq 0$$

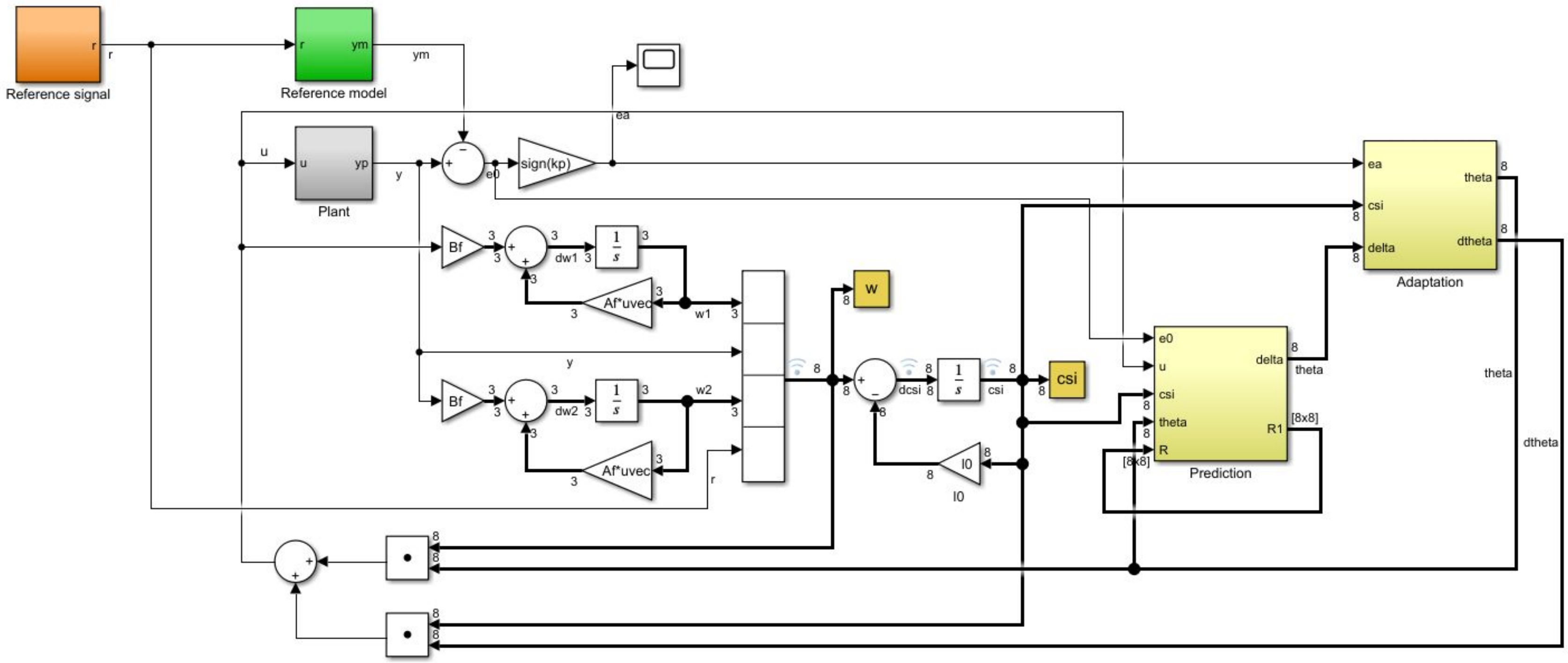
M-MRAC + LS

Resumo do Algoritmo

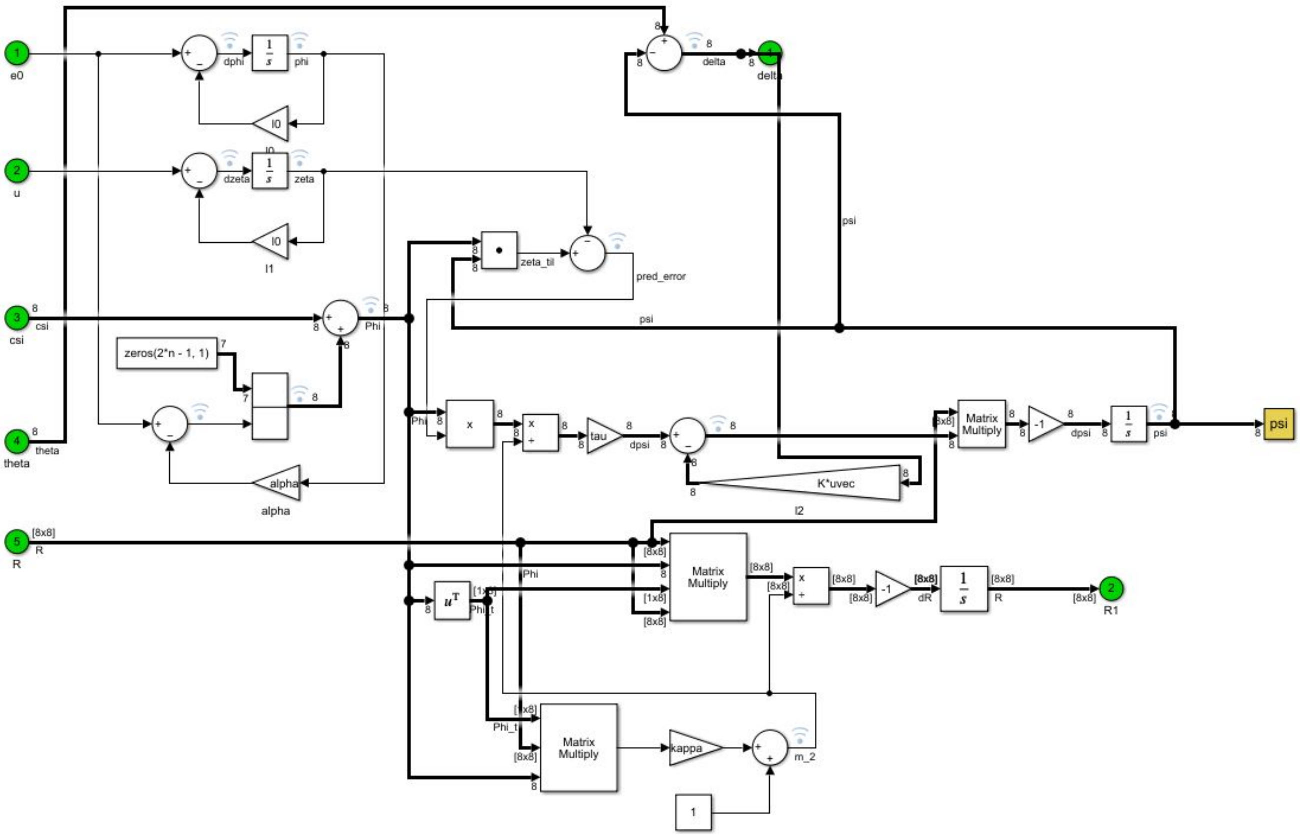
Subsystem	Equation	Order
Plant	$y = P(s)\, u$	n
Model	$y_m = M(s)\, r$	n
Track. error	$e_a = \text{sign}(k_p)(y - y_m)$	
SV-filters	$\dot{\omega}_1 = A_f \omega_1 + b_f u$ $\dot{\omega}_2 = A_f \omega_2 + b_f y$	$n-1$ $n-1$
Regressor	$\omega^T = [\omega_1^T \;\; y \;\; \omega_2^T \;\; r]$	
ξ-filter	$\dot{\xi} = -\ell_0 \xi + \omega$ $\ell_0 > a_m$	$2n$
Control	$u = \theta^T \omega + \dot{\theta}^T \xi$	

Update law	$\dot{\theta} = -\Gamma\, \xi e_a - \sigma \Delta,$ $\Gamma = \Gamma^T > 0, \quad \sigma > 0$ $\Delta = \theta - \psi$	$2n$
Filters	$\dot{\zeta} = -\ell_0 \zeta + u$ $\dot{\varphi} = -\ell_0 \varphi + e_0$	1 1
Prediction	$\hat{\zeta} = \psi^T \phi$ $\phi = \xi + \left[\mathbf{0}^T \;\; (e_0 - \alpha\varphi)\right]^T$ $\mathbf{0} \in \mathbb{R}^{2n-1}$	
Pred. error	$\varepsilon = \hat{\zeta} - \zeta$	
LS estimator	$\dot{\psi} = -R\left(\frac{\tau \phi \varepsilon}{m^2} - \frac{\sigma}{\beta} \Gamma^{-1} \Delta\right)$ $\tau > \frac{1}{2}, \quad \beta > 0$ $\dot{R} = -\frac{R \phi \phi^T R}{m^2}$ $R(0) = R^T(0) > 0$ $m^2 = 1 + \kappa \phi^T R \phi, \;\; \kappa \geq 0$	$2n$ $4n^2$

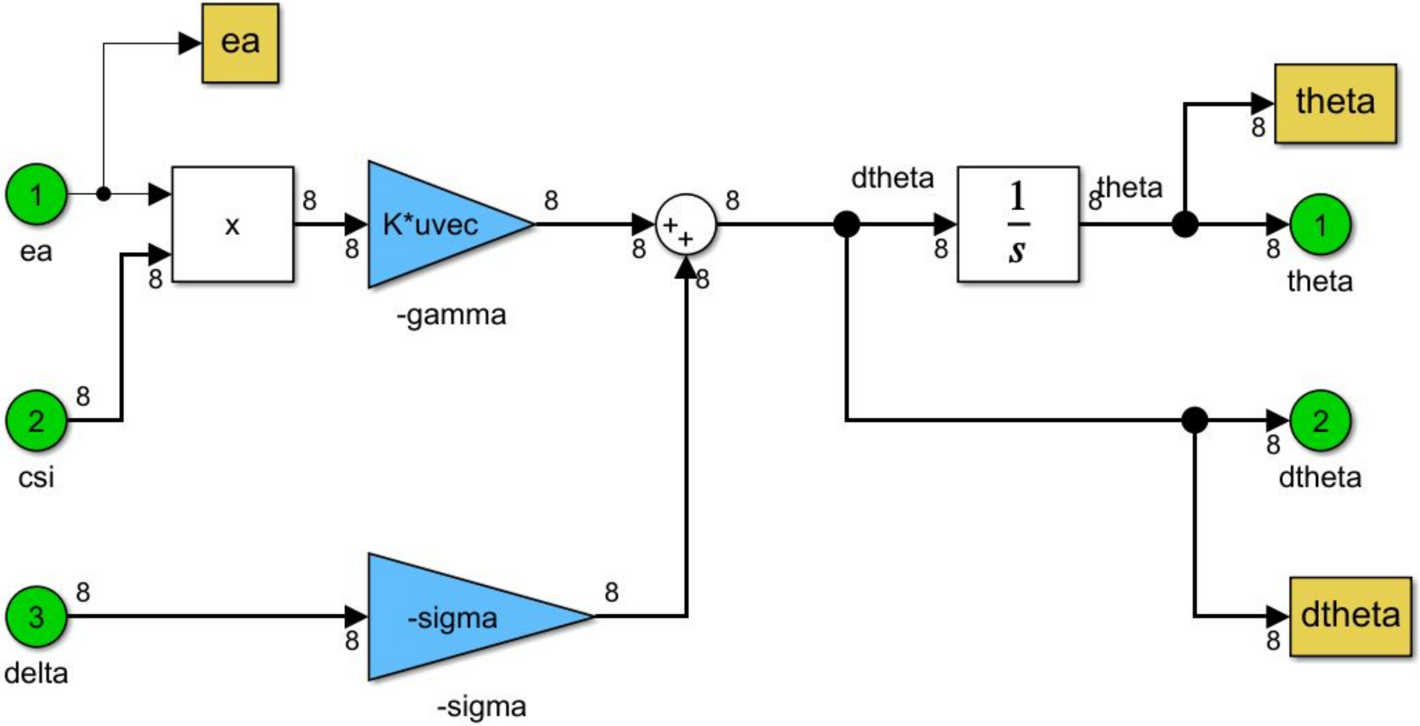
Diagramas de Blocos



Diagramas de Blocos



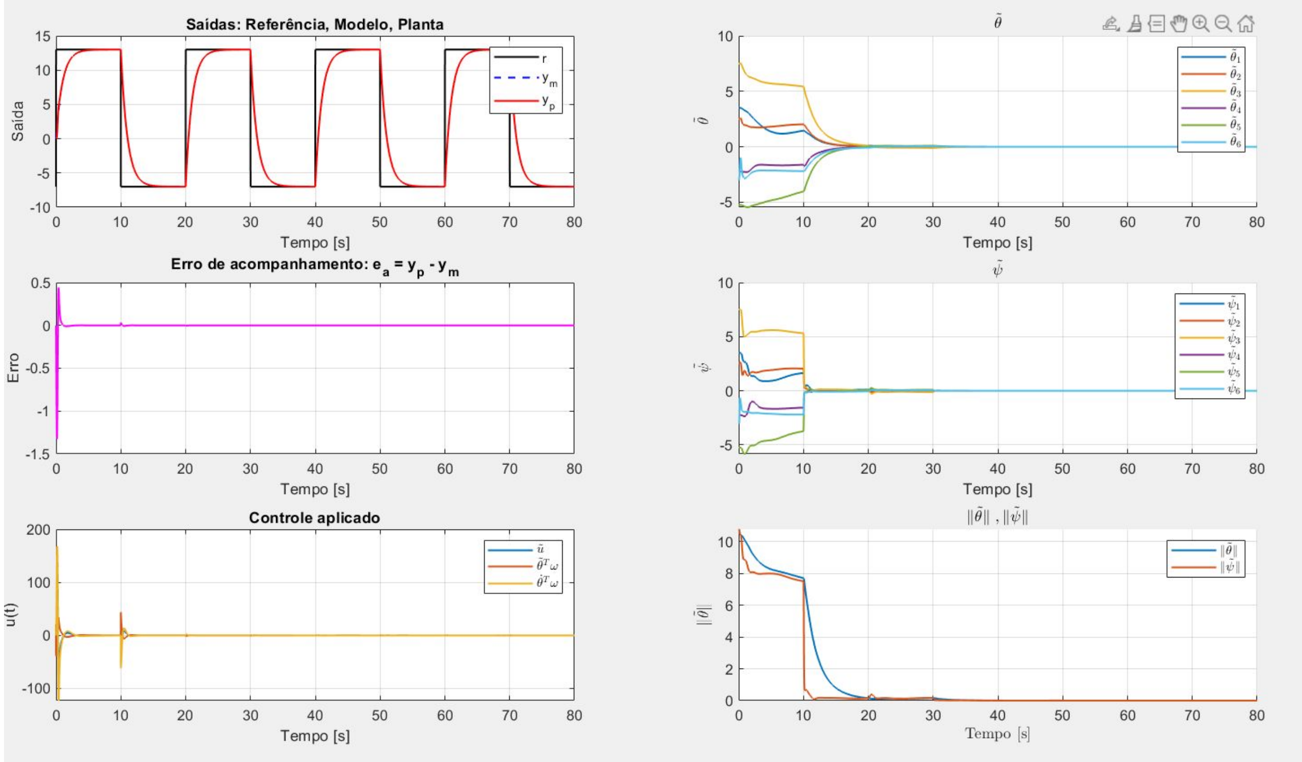
Diagramas de Blocos



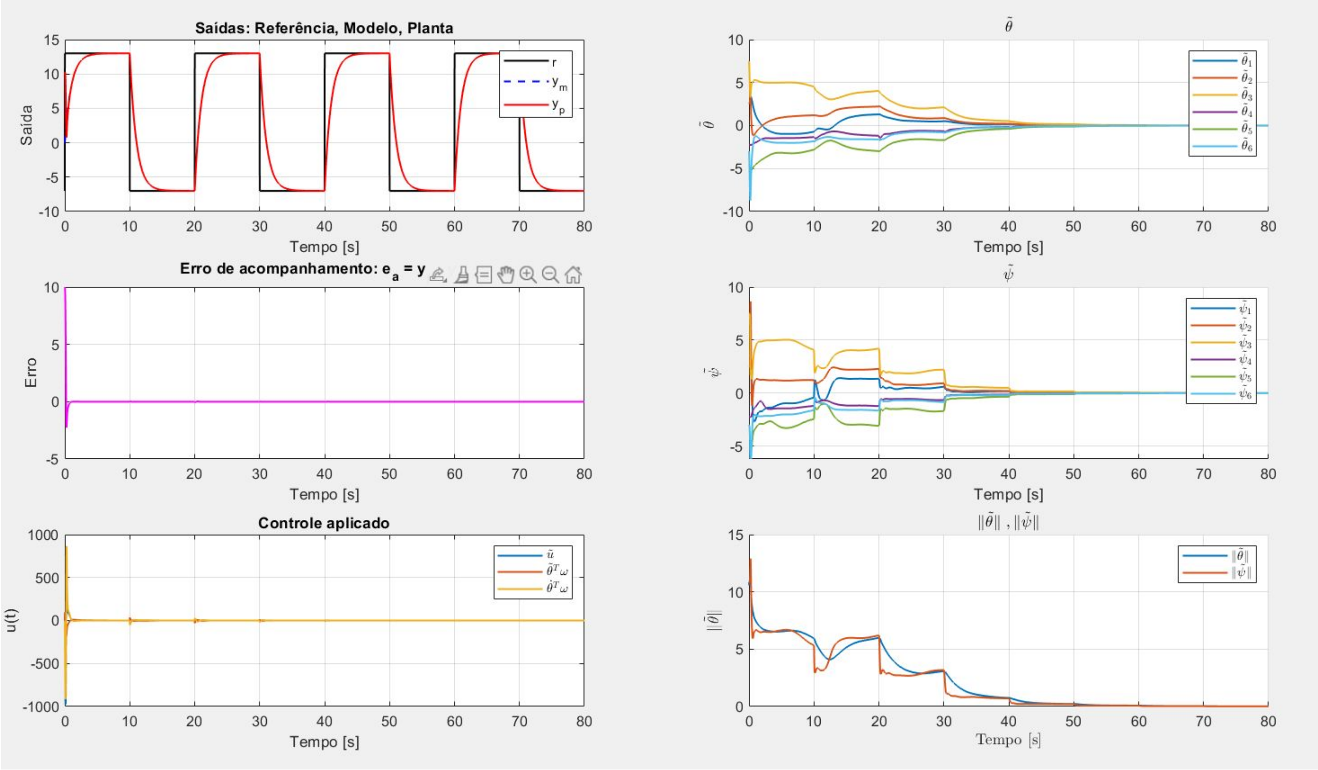
Resultados das Simulações

Base - DC=3 SQ=1

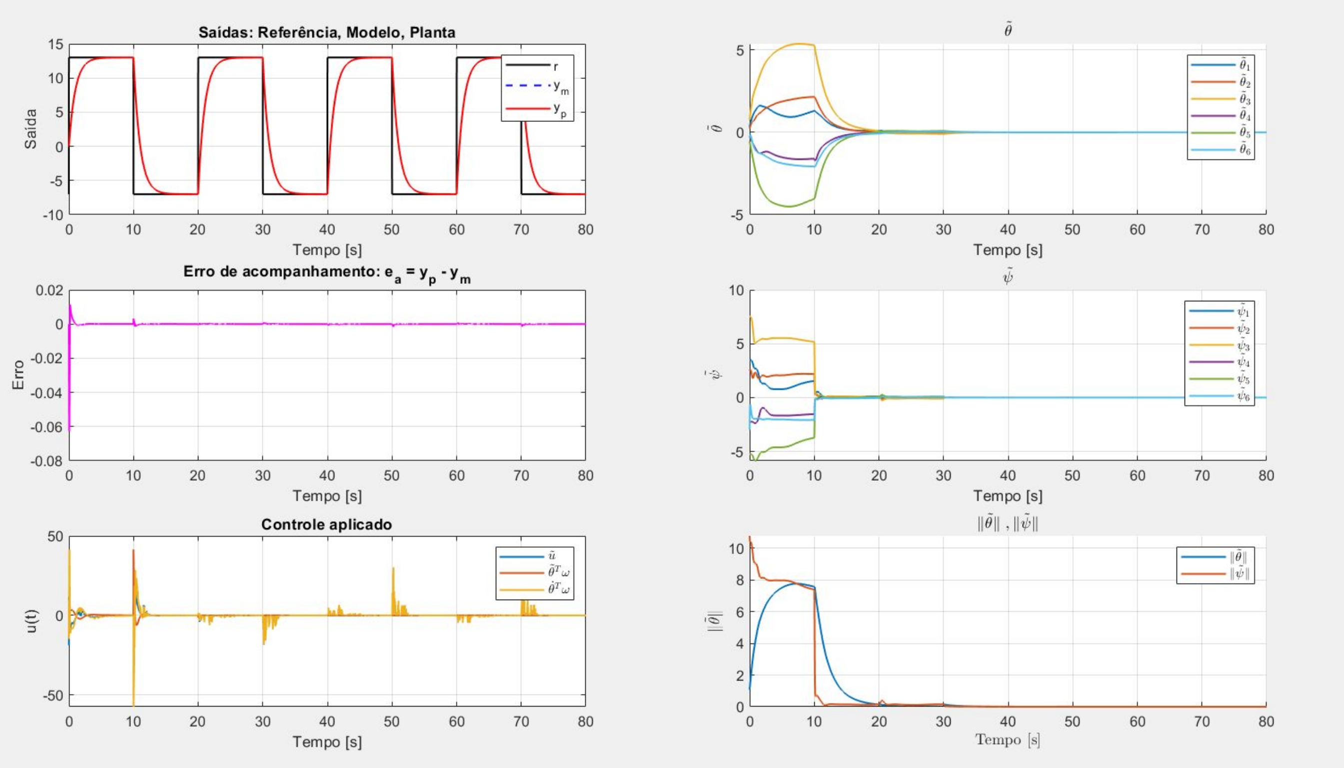
```
n = 3;  % ordem da planta e do modelo

%% ======= Inicialization =======
y0 = 0;
ym0 = 0;
gamma = 5*eye(2*n);
R0 = 200*eye(2*n);
tau=5;
beta=1000;
kappa=0.1;
sigma=0.5;
alpha=1;
tfinal = 80;    %Simulation interval
st = 0.01;      %Sample time to workspace
theta0 = zeros(2*n, 1);

%% ======= Reference signal parameters =======
DC = 3;   %Constant
S1 = 0;
S2 = 0;
S3 = 0;
SQ = 1;
% Define Laplace variable
s = tf('s');
kp= 1/3;
P = kp*((s + 2)^2) / (s^3);
P = ss(P);
x0 = P.c \ y0;
M = 1 / (s + 1);
M = ss(M);
xm0 = M.c \ ym0;
Lambda = 1 / ((s + 0.5)*(s + 1)*(s + 1.5));
l0 = 2;

%% Polinômio do Observador Vetorizado
A0 = 1;
```



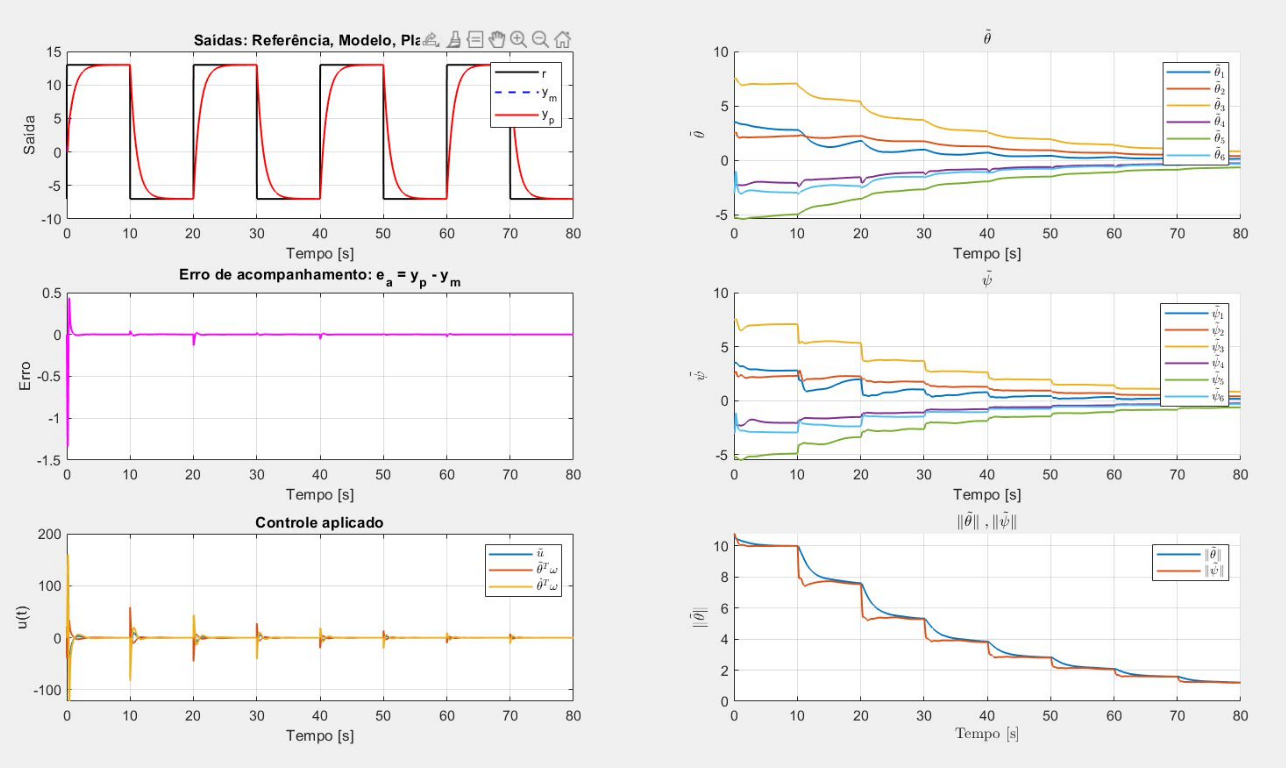
Condições Iniciais - y0 =10



Condições Iniciais - theta = 0.9theta*

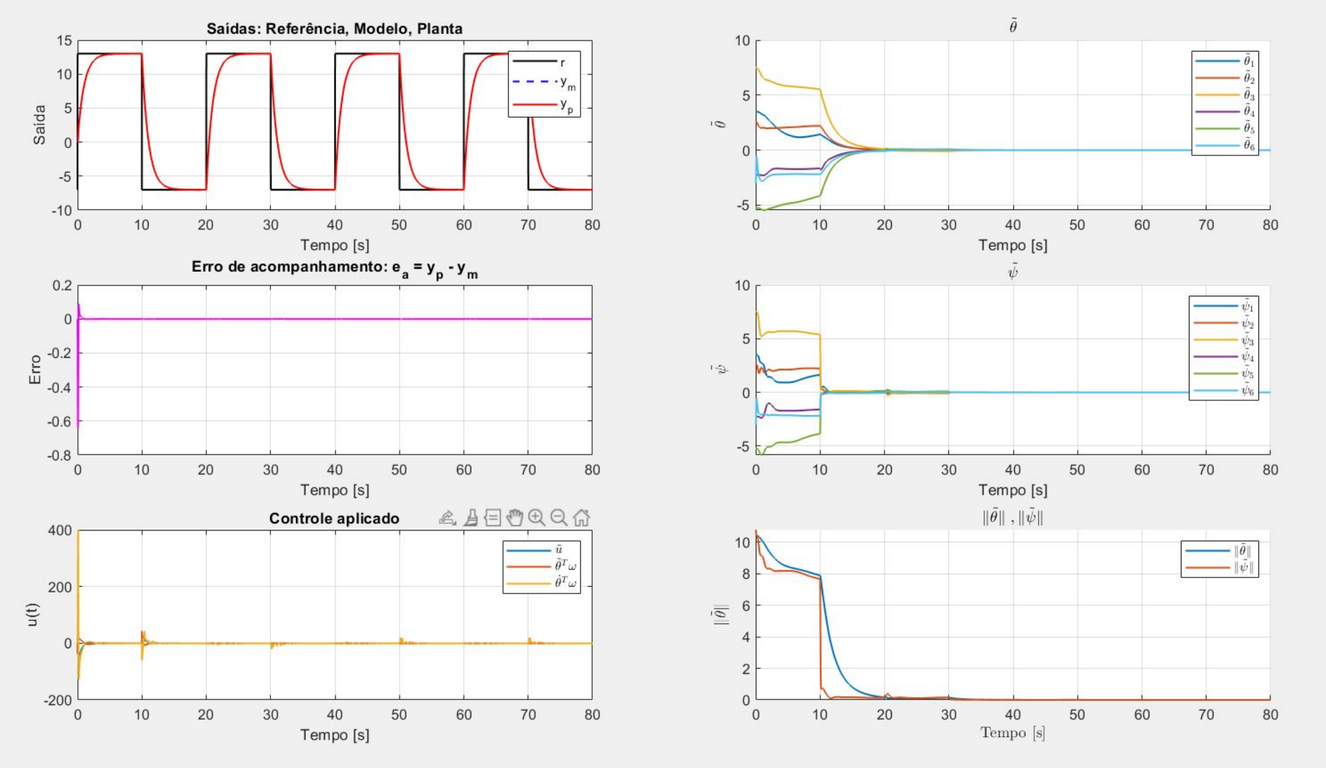


R(0) = 2

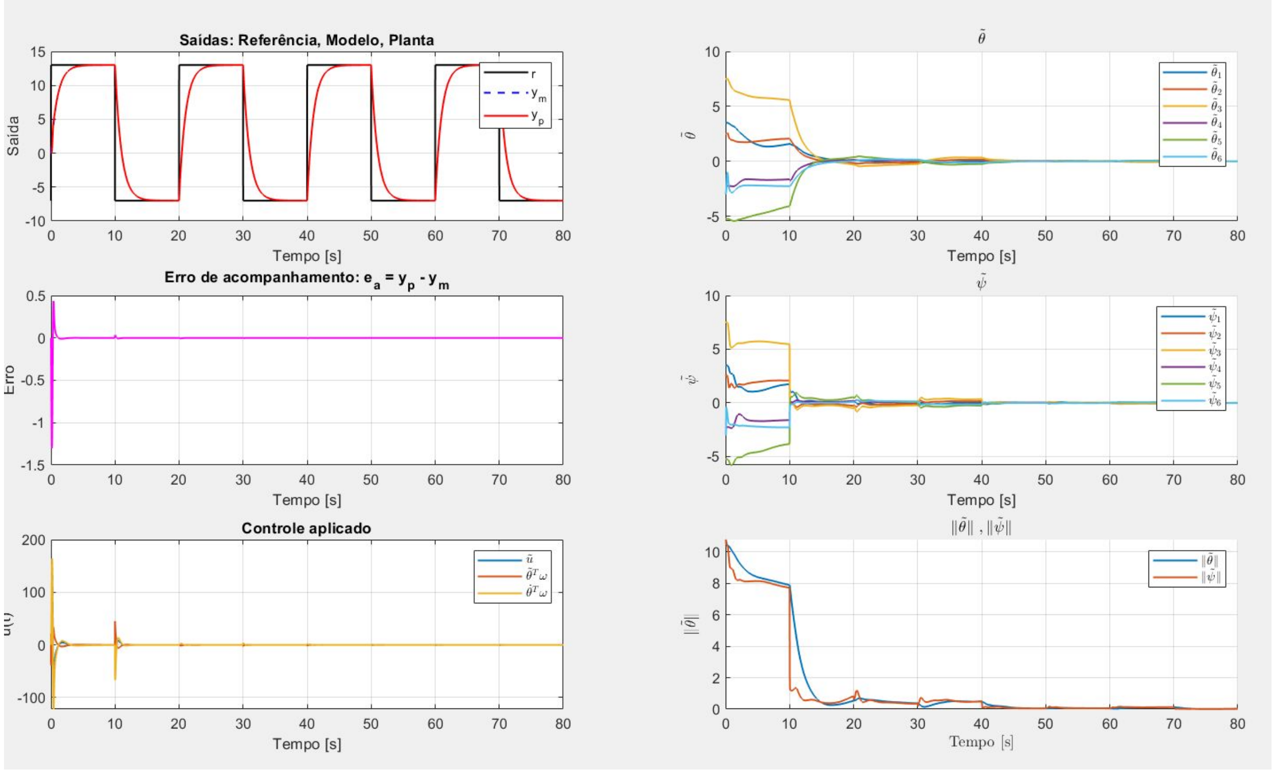


Saídas: Referência, Modelo, Pla
Saída
Tempo [s]
r
y_m
y_p
Erro de acompanhamento: $e_a = y_p - y_m$
Erro
Tempo [s]
Controle aplicado
u(t)
Tempo [s]
$\tilde{u}$
$\tilde{\theta}^T\omega$
$\hat{\theta}^T\omega$
$\tilde{\theta}$
$\tilde{\theta}_1$
$\tilde{\theta}_2$
$\tilde{\theta}_3$
$\tilde{\theta}_4$
$\tilde{\theta}_5$
$\tilde{\theta}_6$
Tempo [s]
$\tilde{\psi}$
$\tilde{\psi}_1$
$\tilde{\psi}_2$
$\tilde{\psi}_3$
$\tilde{\psi}_4$
$\tilde{\psi}_5$
$\tilde{\psi}_6$
Tempo [s]
$\|\tilde{\theta}\|$, $\|\tilde{\psi}\|$
$\|\tilde{\theta}\|$
$\|\tilde{\psi}\|$
Tempo [s]

Ganho de Adaptação (Gamma)= 50

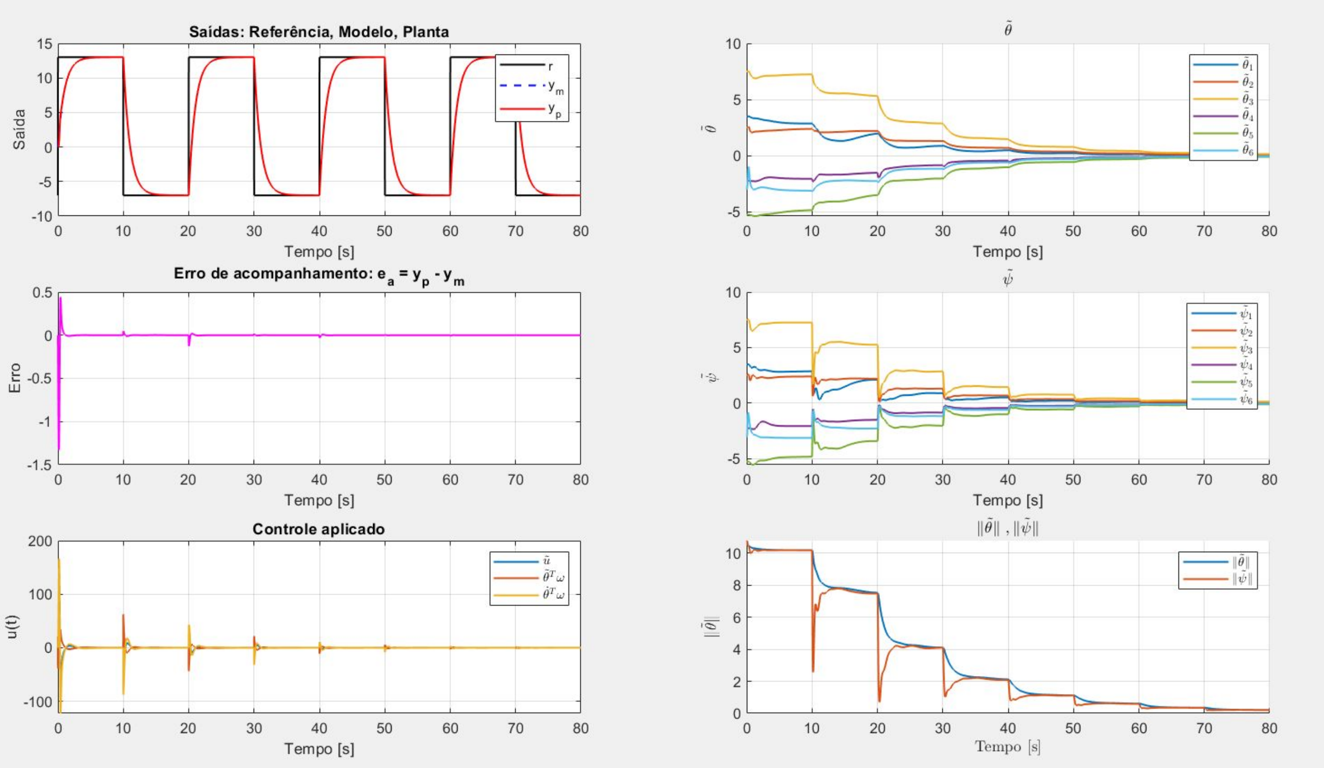


Tau = 50



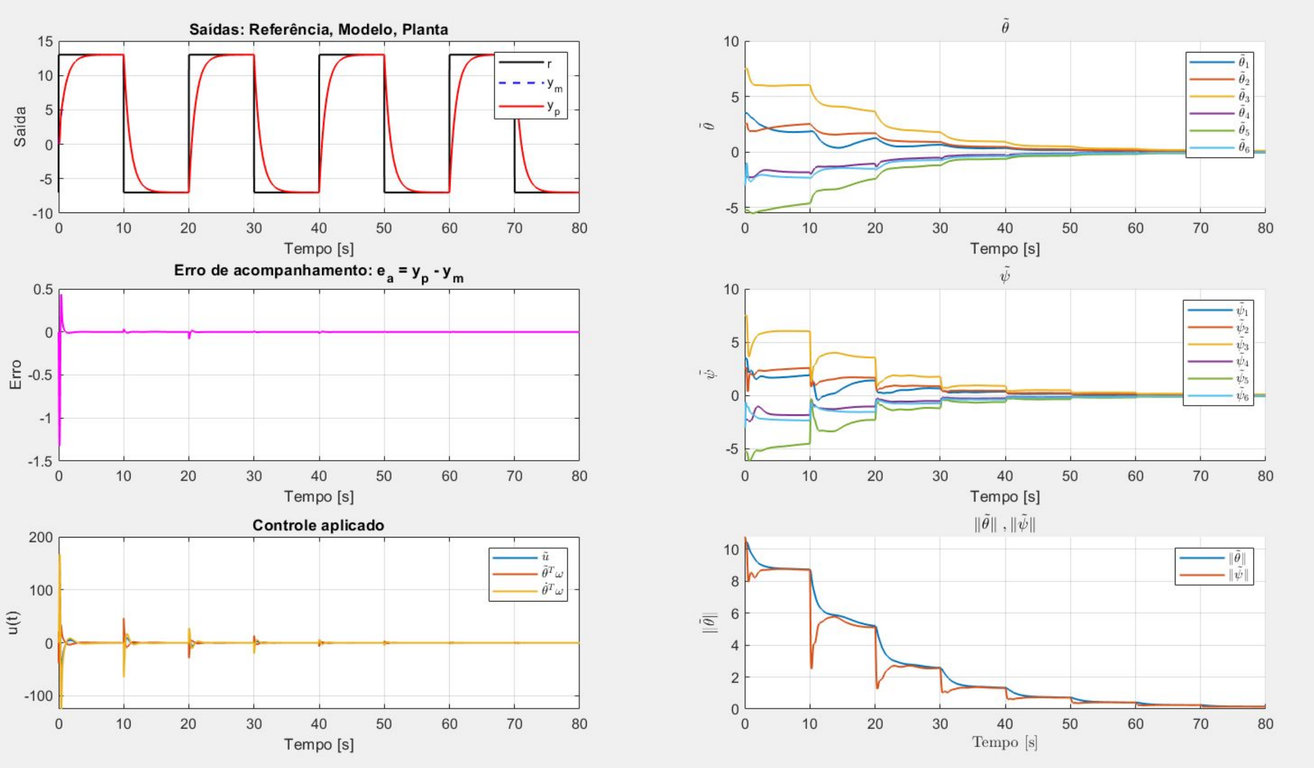
Saídas: Referência, Modelo, Planta
Saída
Tempo [s]
r
y_m
y_p
Erro de acompanhamento: $e_a = y_p - y_m$
Erro
Tempo [s]
Controle aplicado
u(t)
Tempo [s]
$\tilde{u}$
$\tilde{\theta}^T\omega$
$\hat{\theta}^T\omega$
$\tilde{\theta}$
Tempo [s]
$\tilde{\theta}_1$ $\tilde{\theta}_2$ $\tilde{\theta}_3$ $\tilde{\theta}_4$ $\tilde{\theta}_5$ $\tilde{\theta}_6$
$\tilde{\psi}$
Tempo [s]
$\tilde{\psi}_1$ $\tilde{\psi}_2$ $\tilde{\psi}_3$ $\tilde{\psi}_4$ $\tilde{\psi}_5$ $\tilde{\psi}_6$
$\|\tilde{\theta}\|$, $\|\tilde{\psi}\|$
Tempo [s]

Beta = 1



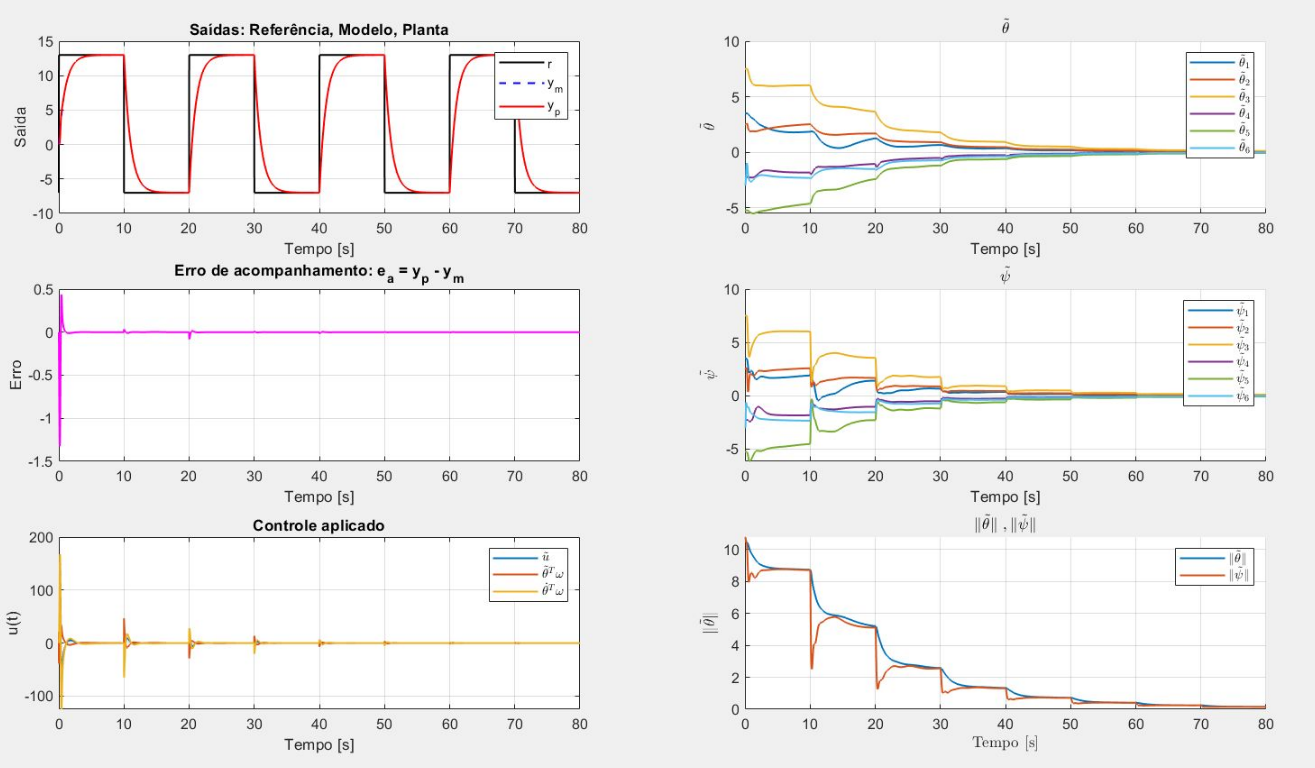
Saídas: Referência, Modelo, Planta
Saída
Tempo [s]
r
y_m
y_p
Erro de acompanhamento: $e_a = y_p - y_m$
Erro
Tempo [s]
Controle aplicado
u(t)
Tempo [s]
$\tilde{u}$
$\tilde{\theta}^T\omega$
$\dot{\theta}^T\omega$
$\tilde{\theta}$
Tempo [s]
$\tilde{\psi}$
Tempo [s]
$\|\tilde{\theta}\|$, $\|\tilde{\psi}\|$
$\|\tilde{\theta}\|$
Tempo [s]

Kappa = 0.01



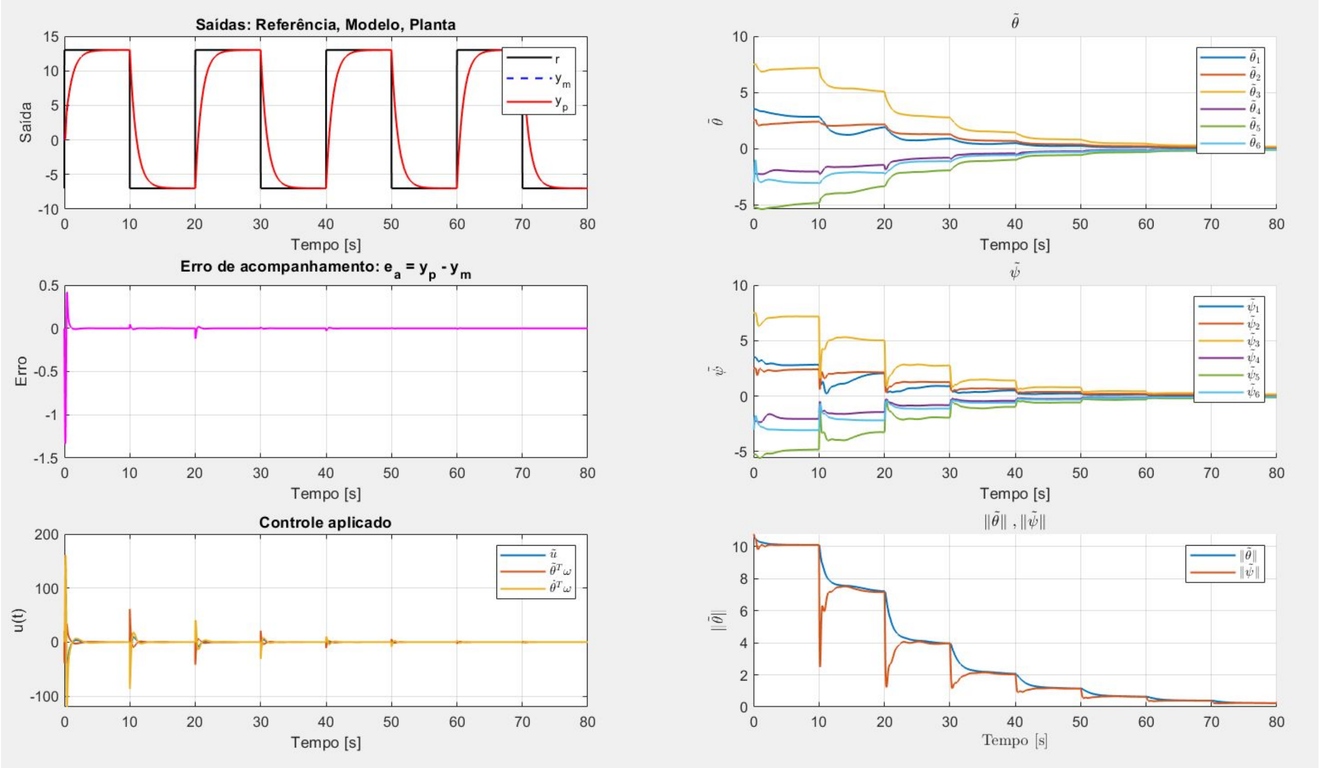
Saídas: Referência, Modelo, Planta
Saída
Tempo [s]
r
y_m
y_p
Erro de acompanhamento: $e_a = y_p - y_m$
Erro
Tempo [s]
Controle aplicado
u(t)
Tempo [s]
$\tilde{u}$
$\tilde{\theta}^T\omega$
$\dot{\theta}^T\omega$
$\tilde{\theta}$
$\tilde{\theta}_1$
$\tilde{\theta}_2$
$\tilde{\theta}_3$
$\tilde{\theta}_4$
$\tilde{\theta}_5$
$\tilde{\theta}_6$
Tempo [s]
$\tilde{\psi}$
$\tilde{\psi}_1$
$\tilde{\psi}_2$
$\tilde{\psi}_3$
$\tilde{\psi}_4$
$\tilde{\psi}_5$
$\tilde{\psi}_6$
Tempo [s]
$\|\tilde{\theta}\|$, $\|\tilde{\psi}\|$
$\|\tilde{\theta}\|$
$\|\tilde{\psi}\|$
Tempo [s]

Sigma = 0.05



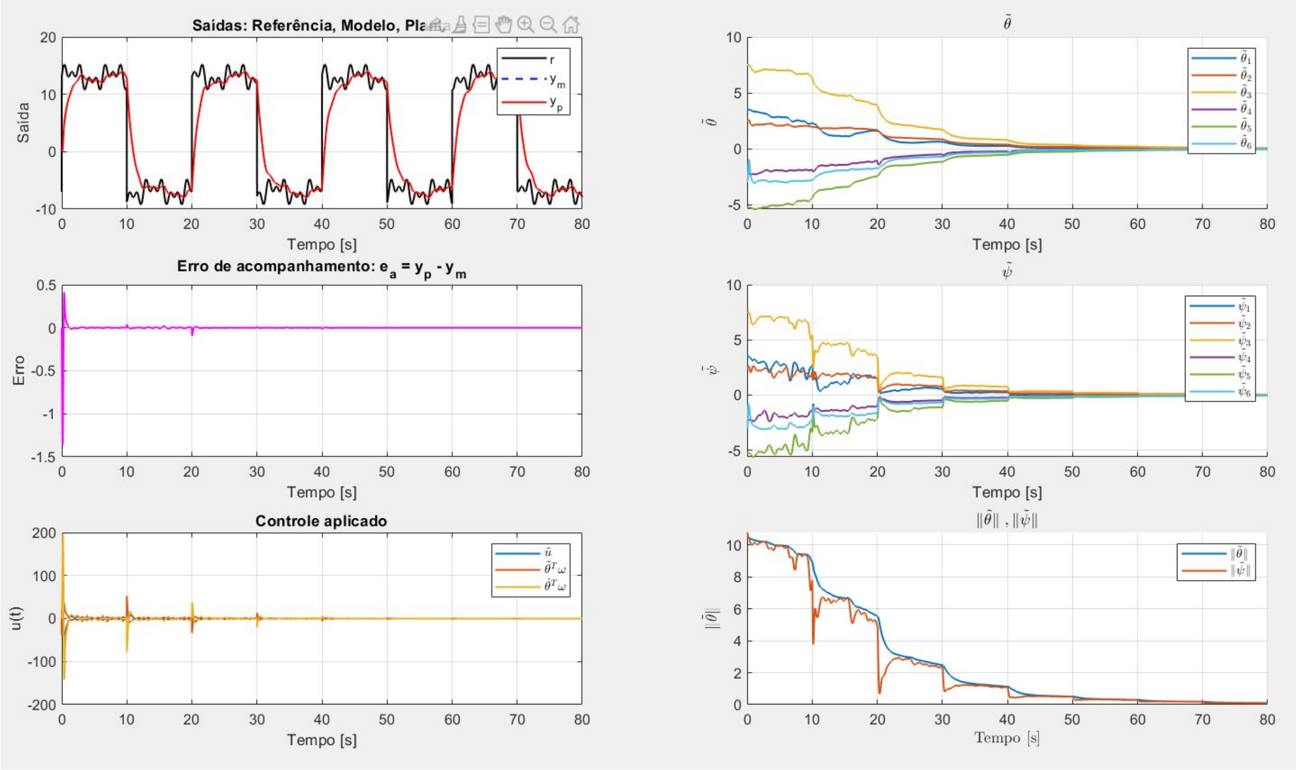
Saídas: Referência, Modelo, Planta
Saída
Tempo [s]
r
y_m
y_p
Erro de acompanhamento: $e_a = y_p - y_m$
Erro
Tempo [s]
Controle aplicado
u(t)
Tempo [s]
$\tilde{u}$
$\tilde{\theta}^T\omega$
$\dot{\theta}^T\omega$
$\tilde{\theta}$
$\tilde{\theta}_1$
$\tilde{\theta}_2$
$\tilde{\theta}_3$
$\tilde{\theta}_4$
$\tilde{\theta}_5$
$\tilde{\theta}_6$
Tempo [s]
$\tilde{\psi}$
$\tilde{\psi}_1$
$\tilde{\psi}_2$
$\tilde{\psi}_3$
$\tilde{\psi}_4$
$\tilde{\psi}_5$
$\tilde{\psi}_6$
Tempo [s]
$\|\tilde{\theta}\|$, $\|\tilde{\psi}\|$
$\|\tilde{\theta}\|$
$\|\tilde{\psi}\|$
Tempo [s]

Alpha = 10



Saídas: Referência, Modelo, Planta
Saída
Tempo [s]
r
y_m
y_p
Erro de acompanhamento: $e_a = y_p - y_m$
Erro
Tempo [s]
Controle aplicado
u(t)
Tempo [s]
$\tilde{u}$
$\tilde{\theta}^T\omega$
$\hat{\theta}^T\omega$
$\tilde{\theta}$
Tempo [s]
$\tilde{\theta}_1$
$\tilde{\theta}_2$
$\tilde{\theta}_3$
$\tilde{\theta}_4$
$\tilde{\theta}_5$
$\tilde{\theta}_6$
$\tilde{\psi}$
Tempo [s]
$\tilde{\psi}_1$
$\tilde{\psi}_2$
$\tilde{\psi}_3$
$\tilde{\psi}_4$
$\tilde{\psi}_5$
$\tilde{\psi}_6$
$\|\tilde{\theta}\|$, $\|\tilde{\psi}\|$
$\|\tilde{\theta}\|$
$\|\tilde{\psi}\|$
Tempo [s]

Sinal de Referência = S1,2,3 = 1; w = 1/3/5 rad/s



Obrigado pela atenção de todos!!!